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Lesson Notes

MODULE 4 | LESSON 2

PROSPECT THEORY

Reading Time	90 minutes
Prior Knowledge	Students are expected to be familiar with the contents of Lesson 1 of Module 4 in which behavioral finance is introduced and the purpose of prospect theory highlighted.
Keywords	Value function and weighting function. Preference for skewness, Bowman's paradox, myopic loss aversion.

The previous lesson ushered us in the world of behavioral finance and showed several key differences with respect to traditional finance. Prospect theory is the key tool for decision making, and an introduction to it was provided.

In the next lesson, we will apply the tenets of behavioral finance and prospect theory to portfolio theory.

1. Prospect Theory in a Formal Manner

In the previous lesson, behavioral finance was ushered in by assuming bounded rationality. Further, we introduced the concept of prospect theory in which risk aversion is replaced by loss aversion since Kahneman and Tversky, based on experimental findings, taught us that economic agents fear losses more than they value gains.

Now, the question becomes the following: Is prospect theory useful for analyzing investment decisions, i.e., decisions that deal with risky outcomes?

To answer this question, we need to understand how prospect theory works when individuals need to come up with decisions/choices.

Prospect theory consists of two phases:

- **Editing phase:** aimed at reformulating and/or reorganizing the available options of the problem to the end of simplifying the decision making process. It is implemented via several operations, which will be explained below.

- **Evaluation phase:** the decision maker is expected to evaluate each of the edited options and choose the prospect with the higher value. As explained by Kahneman and Tversky, the overall values of the edited prospects, which we choose to denote by V , are expressed in terms of two scales. The first scale associates to each physical probability, p , a decision weight, $w(p)$. The second scale v assigns to each outcome x a number $v(x)$, which can be regarded as a subjective value of the outcome x . In particular, we stress that the outcomes are defined relative to a reference point, and thus, V measures the values of deviations from the reference point or, in other words, the value of specific wins or losses. We have already studied this part in the previous lesson.

1.1 Editing Phase

The editing phase involves several steps or, more properly, operations that transform the outcomes and the probabilities related to the prospects. The steps are:

- **Coding:** each proposal is coded as a gain or a loss of value relative to a reference point. Usually, but it is not strictly necessary, the reference point corresponds to the current value of the asset that is involved. Furthermore, probabilities are assigned to each possible event. Note that the values and losses can be different for different individuals.
- **Combination:** used to simplify the prospects by combining the probabilities of outcomes with identical values.
- **Segregation:** aimed at separating expected returns into risky and risk-free components (if possible, of course).
- **Cancellation:** removing any outcomes common to two or more proposals does not affect the decision.
- **Simplification:** described as the process consisting in rounding the probabilities of the outcomes of a prospect in the sense that if they differ by very small amounts or the outcomes are very unlikely, we can consider the probabilities the same or zero, respectively.
- **Detection of dominance:** we could discard from consideration any alternative that is clearly dominated and eliminate it from any further evaluation.

Examples are necessary to illustrate each of these entries and make them clearer.

Example 1. Consider a portfolio that consists of four assets: A, B, C, and D. Let's assume that we have the following table of information:

Asset	Current price	Historical price (cost basis)	Close price, yesterday
A	10	8	11
B	13	13	14
C	15	12	16
D	19	12	20

And suppose that we want to detect which asset suffered the largest percentage loss. To do this, we need first to decide which reference points to use: Is the reference point yesterday's close, or is it cost basis? Or perhaps we should use the year-end closing prices? If we used the cost basis, then the percentage change in prices would be:

+25%, 0%, 25%, 58.33%

in which case the asset with the largest percentage loss would be B with 0%. If we used yesterday's close, then the percent changes in price would become:

−9.09%, −7.14%, −6.25%, −5%

and we would select asset A. Hence, a reference point is necessary, but what we performed after that is what Kahneman and Tversky described as coding.

Example 2. Again, we describe the situation with a table:

Outcomes	probability of outcomes	Expected return for that outcome
x1	0.09	-10%
x2	0.15	15%
x3	0.15	8%
x4	0.25	15%
x5	0.15	15%
x6	0.21	20%

This table displays three outcomes as having the same expected value (0.15), and they are x2, x4, and x5. Using combination, this table can be simplified as follows:

Outcomes	probability of outcomes	Expected return for that outcome
x1	0.09	-10%
x3	0.15	8%
x2, x4, x5	0.55	15%
x6	0.21	20%

So, combination has allowed us to simplify our table.

Example 3. Suppose that a gamble offers the following payoffs:

outcome	probability	payoff
x1	0.60	\$100
x2	0.40	\$130

This gamble is the same as the following:

outcome	probability	payoff
x3	1.00	\$100
x4	0.40	\$30

In other words, by using segregation, we were able to reduce the original gamble to a new one in which the gambler earns \$100 with probability 1 (riskless) and \$30 with probability 0.40.

Example 4. Consider two proposals and their expected values and probabilities as described in the following table:

Proposal A		
outcome	probability	payoff
x1	1/3	10
x2	1/3	20
x3	1/3	30

Proposal B		
outcome	probability	payoff
x1	1/2	10
x2	0	20
x3	1/2	30

If we use cancellation, the outcomes that are common to both proposals can be removed and we would get the following tables instead:

Proposal A		
outcome	probability	payoff
x1		
x2	1/3	20
x3		

Proposal B		
outcome	probability	payoff
x1	1/6	10
x2		
x3	1/6	30

which is simpler to use.

Example 5. Let's say that a gamble has the following payoffs and probabilities:

outcome	probability	payoff
x1	0.49	\$500
x2	0.50	\$800
x3	0.01	\$1000

Simplification can be used, and it yields the following table instead:

outcome	probability	payoff
x1	0.50	\$500
x2	0.50	\$800
x3	0.00	\$1000

Example 6. Detection of dominance is quite simple. Consider two proposals, A and B, whose payoffs and probabilities are summarized by the following table:

Proposal A		Proposal B		
outcome	probability	payoff	probability	payoff
x1	0.2	10	0.2	5
x2	0.5	20	0.5	15
x3	0.3	30	0.3	26

Since for each outcome the payoff of proposal A is higher than the equivalent payoff of proposal B and the probabilities of the outcomes are the same we can say that proposal A dominates proposal B (higher average, higher minimum, higher maximum). Hence, proposal B can be discarded.

1.2 Evaluation Phase

In lesson 1 of this module, we have already seen how to evaluate prospects. Here, we will be a bit more formal. Once the editing phase is completed, the decision maker is expected to evaluate each of the edited prospects and choose the prospect with the highest value. Kahneman and Tversky (299) explain "loss aversion, risk seeking, and nonlinear preferences in terms of the value and the weighting functions."

Hence, if we denote by V the overall value of an edited prospect, V can be expressed in terms of two scales: w and v .

- The first scale associates to each physical probability, p , a decision weight $w(p)$ which reflects the effect of the probability on the total value of the prospect.
- The scale v assigns a number to each outcome x , $v(x)$. The value v is subjective, and it measures the value of deviations (gains or losses) from the reference point.

Thus, given a set of possible outcomes

$$V(x_1, \dots, x_n) = p_1 \cdot w(p_1) \cdot v(x_1) + p_2 \cdot w(p_2) \cdot v(x_2) + \dots + p_n \cdot w(p_n) \cdot v(x_n).$$

the "utility" function in prospect theory looks like:

$$V(x_1, \dots, x_n) = w(p_1) \cdot v(x_1) + w(p_2) \cdot v(x_2) + \dots + w(p_n) \cdot v(x_n).$$

Example 7. Suppose that we are evaluating a prospect. Without loss of generality, We can assume that the lowest probability weighted payoff is achieved for the outcome x_1 (if that is not the case,

we can always relabel the outcomes so that the outcome with the lowest weighted payoff $w(p_i) \cdot v(x_i)$ is achieved for $i = 1$). Then, during the editing phase, segregation allows us to write.

$$V(x_1, \dots, x_n) = w(p_1)v(x_1) + \sum_{i=2}^n p_i[(w)p_i(v(x_i) - w(p_1)v(x_1))]$$

Hence, in this case, we have a riskless component $w(p_1) \cdot v(x_1)$ which is the minimum gain or loss and a risky component that depends on the other outcomes and their probabilities.

We need to spend a bit more effort on the value function. We have justified an asymmetric S shape for it, but we can say more. For instance, Kahneman and Tversky in their 1992 paper proposed the following function

$$v(x) = \begin{cases} x^\alpha & \text{if } x \geq 0; \\ -\lambda(-x)^\beta & \text{if } x < 0 \end{cases}$$

and estimated the parameters as $\alpha, \beta \approx 0.88$ and $\lambda \approx 2.25$. The fact that $\alpha \approx \beta = 0.88$ informs us that there is 1.) a diminishing sensitivity to both positive and negative extreme outcomes, and 2.) substantial loss aversion since $\lambda = 2.25$ describes the slope of the loss function vs. the gain function. In practical terms, the disutility of giving something up is more than two times the utility of gaining the same amount.

Another approach exploits cumulative distribution functions. In particular, using this approach, we can obtain a parametric version of the value function. This is useful as we can adjust it to different instances. To see how this works, let Y be a random variable that takes only nonnegative values and assume that $Z = \log(Y)$ follows a logistic distribution. Z is known as a log-logistic distribution. This means that Z has a CDF that looks like this

$$F(z, \mu, \sigma) = \frac{1}{1 + (z/\mu)^{-\sigma}}.$$

By multiplying both numerator and denominator by $(x/\mu)^\sigma$ and some other algebraic manipulations, it is possible to rewrite

$$F(z, \mu, \sigma) = \frac{z^\sigma}{\mu^\sigma + z^\sigma}.$$

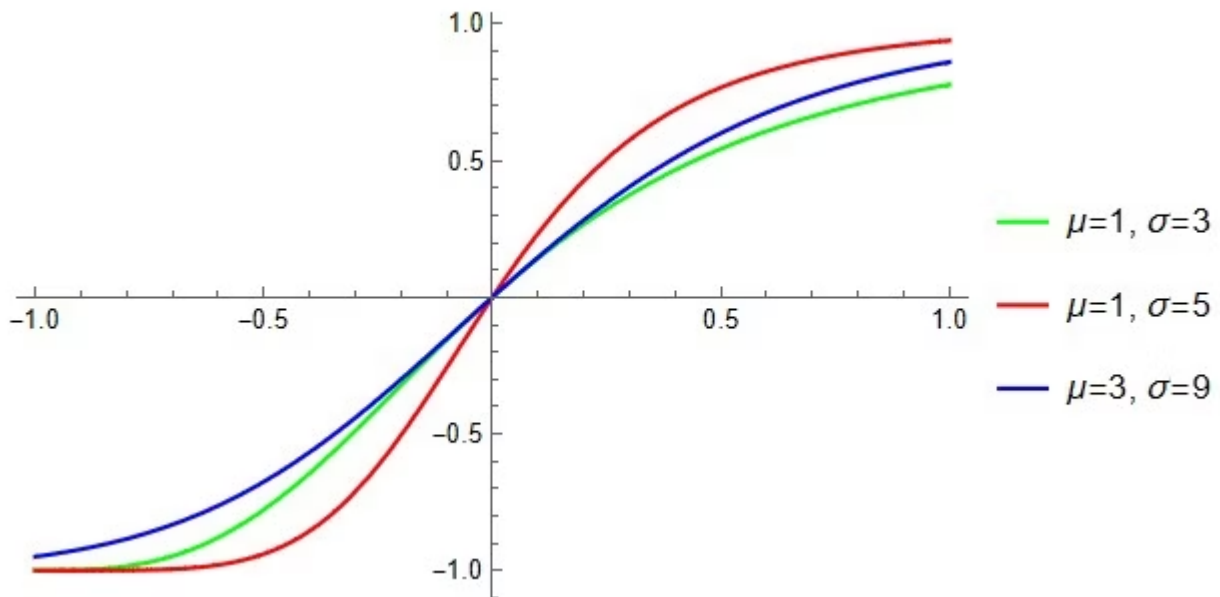
Now, by changing the values of the parameters μ and σ , we can obtain many different versions of the value function. This function is not completely satisfactory for our purposes yet. However, if we shift it left and down, then we obtain a veritable candidate for the value function:

$$F(z, \mu, \sigma) = \frac{(z + \mu)^\sigma}{\mu^\sigma + (z + \mu)^\sigma} - \frac{1}{2}.$$

Finally if we multiply it by 2, we get a v function with asymptotes ± 1 :

$$F(z, \mu, \sigma) = \frac{2(z + \mu)^\sigma}{\mu^\sigma + (z + \mu)^\sigma} - 1.$$

In figure 1, three such value functions have been plotted to see how the choice of the parameters μ and σ affect the overall shape of the value curve.

Figure 1: Value Function Derived from the Log-Logistic Distribution

Another suitable candidate for the value function can be obtained using a lognormal distribution. Although it is not possible to write the CDF of the lognormal distribution in a closed form fashion, it can nevertheless be expressed as

$$F(\ln(x), \alpha, \beta) = \frac{1}{2} + \frac{1}{2} \operatorname{erf}\left(\frac{\ln(x) - \alpha}{\beta\sqrt{2}}\right).$$

with **erf** denoting the error function, i.e.,

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-u^2} du.$$

The log-logistic and the log-normal distributions, once the parameters $\alpha, \beta, \mu, \sigma$ are chosen in a suitable manner, produce very similar value functions and are both very suitable to produce value functions that can be adapted to different real-life situations using mathematical models with just two parameters.

Now, we need to spend a few words about the probability weight function. In the previous lesson, we covered Kahneman and Tversky's motivations. Earlier versions used monotonic transformation of outcomes probability and that did not always produce the desired conclusions theorists had in mind. Kahneman and Tversky's original contribution is seen in the fact that they use a "rank dependent functional" in which one does not transform each probability separately, but the entire cumulative distribution function. In the previous lesson we considered situations in which the random outcome were just a few; however it would be difficult to implement that approach if we had tens and tens of outcomes and even more so if we had a continuous of outcomes.

Empirical studies suggested an inverse S-shaped weighting function, concave below a certain threshold and convex above the same (to be consistent with risk-seeking for low probability gains and risk aversion for high probability gains). Kahneman and Tversky proposed a family of weighting functions that uses two parameters:

$$w^+(p) = \frac{p^\gamma}{[p^\gamma + (1-p)^\gamma]^{1/\gamma}}$$

for the probability of outcomes that lead to gains, and

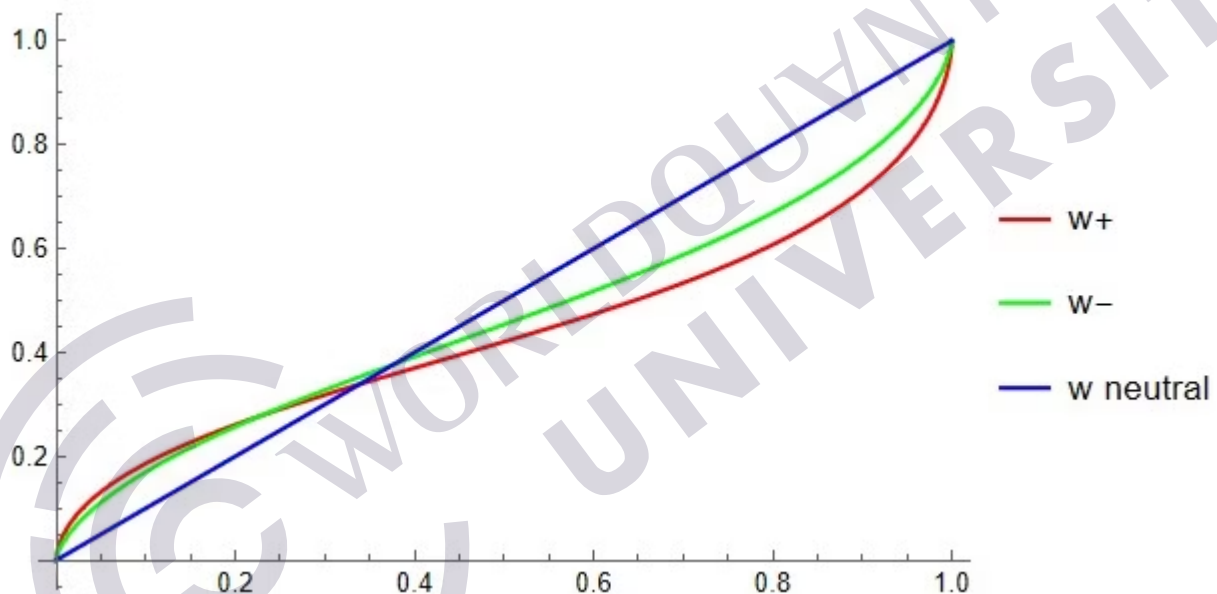
$$w^-(p) = \frac{p^\delta}{[p^\delta + (1-p)^\delta]^{1/\delta}}$$

for the probability of outcomes that lead to losses instead.

By modifying the parameters γ and δ , we can change the "stretching" of the weighting function. The curvature of the weighting function aims at depicting the individuals' attitude towards low, moderate, and high probabilities. In particular, when $\gamma = \delta = 1$, then we get back the original probabilities, i.e., $w^\pm(p) = p$, which we could call the perfectly risk-neutral weighting. In the same 1992 papers we mentioned earlier, Kahneman and Tversky estimated $\gamma = 0.61$ and $\delta = 0.69$.

This particular weighting function is displayed in figure 2. It is clear from the picture that individuals underweight high probabilities both in the case of gains and losses, but more so in the case of gains. Similarly, individuals overweight small probabilities for both gains and losses, but slightly more so in the case of losses.

Figure 2: Kahneman and Tversky Weighting Function



A study by Gurevich et al. using stock option data for the U.S. produced values of $\gamma = 0.84$ and $\delta = 0.76$. This agrees with the result of Kahneman and Tversky, although in this case, the w^+ and w^- transformations are closer to the straight line of risk-neutral weighting.

Remark: The paper by Gurevich et al., while short, does not make for a truly simple reading as it requires the knowledge of stochastic discount factors and state prices, which are usually studied in advanced classes in financial economics. Nevertheless, it is highly recommended for students in quantitative fields. Behavioral finance does not use complex optimization calculations like traditional finance; however, it requires a certain mix of quantitative abilities like estimating the right value and weighting functions. For that reason, the reference to this paper is provided, and it makes for a useful voluntary reading. The mathematics used in the paper is not particularly advanced.

In what follows, we will use prospect theory to explain certain paradoxes or puzzles. This highlights that prospect theory is not just an academic creation for show, but it can actually explain real human behaviors as far as investing is concerned.

2. Preference for Skewness

Individual investors can be optimistic or pessimistic. In some ways, we could say that a positive sentiment for the market leads investors to expect optimistic outcomes while a negative sentiment triggers a pessimistic sentiment about the markets, which leads to expecting negative outcomes. However, the evidence suggests that among individual investors, the difference between pessimistic or optimistic individuals does not lie in how they allocate their portfolios to equities, bonds, and cash, but rather in the amount of diversification that they employ in allocating their portfolios. More specifically, optimists tend to ignore the benefits of diversification—so much so that some authors have asked themselves why optimism tends to have a positive effect on the rate of success of entrepreneurs and self-employed people in general. Nevertheless, it fails to produce a similar outcome among investors where the results are usually not good, if not outright deleterious. In this sense, Statman (44) noticed in a 2004 paper that there is a problem (diversification puzzle):

"The levels of diversification in U.S. investors' equity portfolios present a puzzle. Today's optimal level of diversification, measured by the rules of mean variance portfolio theory, exceeds 300 stocks, but the average investor holds only 3 or 4 stocks."

This happens because:

1. optimists overestimate returns or underestimate risks;
2. pessimists underestimate returns or overestimate risks.

These facts can be revisited in more mathematical terms by stating that optimists have a **preference for skewness**; i.e., they are willing to accept lower returns in exchange for positive skewness and the possibility of very large payouts. This preference for skewness can be incorporated by modifying the weighting function.

Earlier, we used the following piecewise function introduced by Kahneman and Tversky:

$$w^+(p) = \frac{p^\gamma}{[p^\gamma + (1-p)^\gamma]^{1/\gamma}}$$

for the probability of outcomes that lead to gains, and

$$w^-(p) = \frac{p^\delta}{[p^\delta + (1-p)^\delta]^{1/\delta}}$$

for the probability of outcomes that lead to losses, instead with $\gamma = 0.61$ and $\delta = 0.69$. The preference for skewness can be modeled by using a different set of parameters, say $\gamma = 0.4$ and $\delta = 0.6$ (see figures 3 and 4).

Figure 3: Preference for Skewness: Weighting Function for Gains

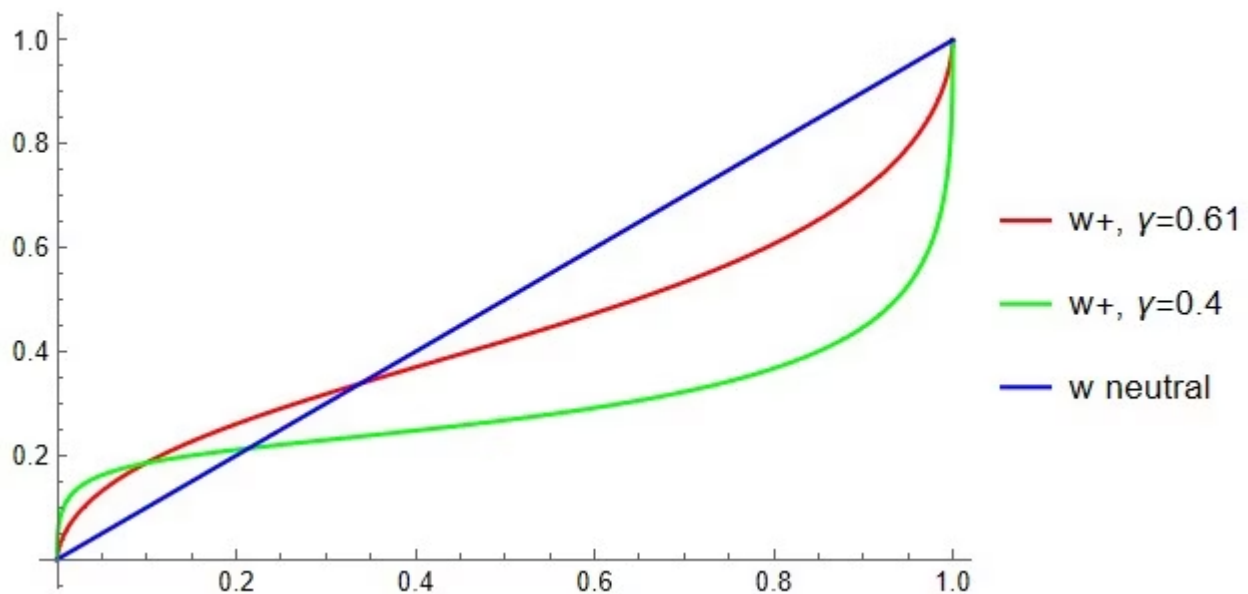
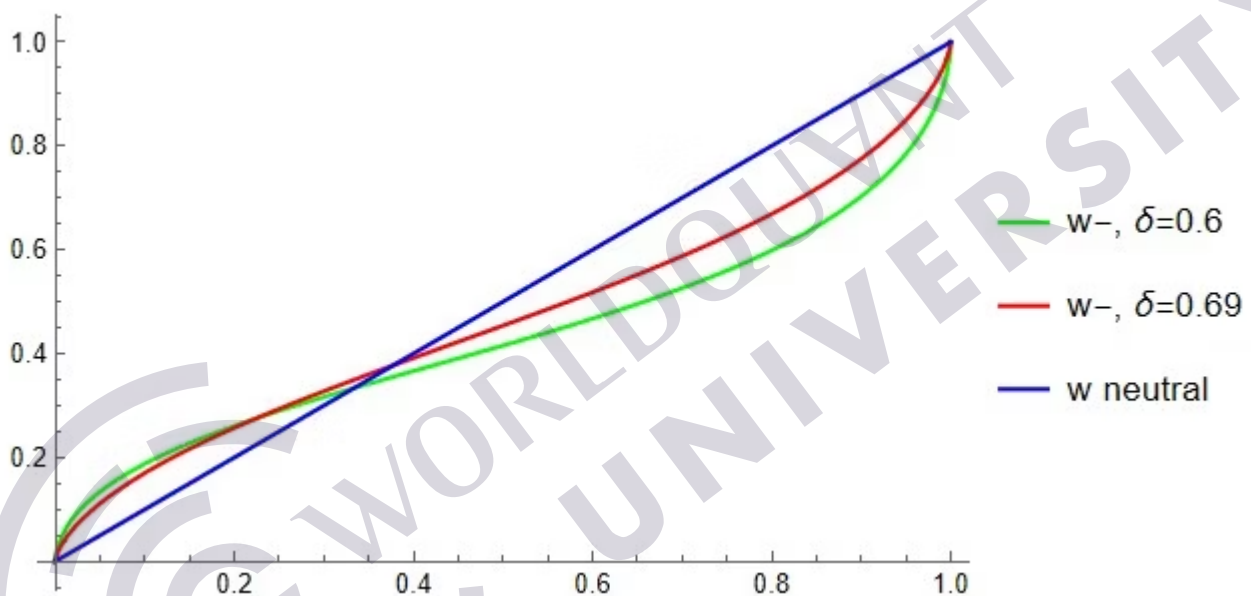


Figure 4: Preference for Skewness: Weighting Function for Losses



In these figures, it is clear that there is a more extreme over-weighting of low probabilities and under-weighting of large probabilities than in the original function used by Kahneman and Tversky.

The relationship we just highlighted between low returns, positive skewness, and investors' preference toward investments that offer lottery-like payoffs is observed across a wide variety of financial settings. One recent trend that witnesses this type of behavior is the increasing participation in lotteries and the like.

3. Bowman's Paradox Explained via Prospect Theory

William J. Bernstein—a neurologist who later became a financial analyst—pointed out that in spite of the general expectation that growth stocks will earn a higher return than value stocks due to the fact that they are usually high beta stocks and thus contain a lot of systematic risk, which makes them overall riskier than value stocks, in reality, historical returns point to the opposite. The fact that low volatility stocks earn higher returns than high volatility stocks poses a serious challenge to the notion of rational expectations and has deserved its own name of volatility anomaly.

Edward H. Bowman observed in 1980 that the riskiest firms provided the lowest returns, a result confirmed in further articles in 1982 and 1984. The difference with what we discussed in the previous paragraph is that Bowman uses a *firm-level accounting-based risk and return and not financial returns*. More specifically, Bowman uses mean values of ROA as a proxy for return and variance of ROA as a proxy for risk. His approach has been criticized on the basis that using moments of the same variable might lead a statistical bias since the moments of any same variable could be positively related to each other. Furthermore, successive studies reported that skewness of return distributions can lead to spurious negative relationships between risk and return. Nevertheless, further studies have not been able to invalidate these findings, not completely in any case. And, in at least one instance, when total debt to total asset ratio is used as a risk measure, significant negative association between performance (measured via ROA or ROE) and leverage for all company size categories were detected. This has become known as **Bowman's paradox** and presents a challenge to the strategic management's tenets that higher risk needs to be compensated by higher returns; otherwise, management would not be motivated to undertake risky investments/projects. In fact, Bowman observed something even stronger, i.e., that high levels of managerial skills produced low returns more often than high returns.

Several attempts have been made to explain Bowman's paradox in the context of TF. Prospect theory has been invoked as well as using the facts that individuals use reference points or targets to evaluate risky choices and individuals are not risk-averse in a uniform manner. In fact, when returns have been below the target, managers can become risk-seekers, and instead, when returns have been above a given target, the majority of managers become risk-averse. In other words, companies may be prone to behave like individuals. Note that the reference point plays a very important role in this formulation. What is the reference point? In this case, it is a suitable target level of return, which could be based on industry-wide averages or some industry-wide benchmark.

Applications of prospect theory to solving the Bowman's paradox have revealed that the expected relationship between risk and return fails to materialize for firms with returns close to the target level—or that in some cases, there is no significant correlation between the two around the threshold level of the return target.

4. Equity Premium Puzzle and Market Participation Puzzle: A Lesson from Prospect Theory

Another situation that has puzzled economists for a while has to do with the equity risk premium. In finance, we are taught that investors do not like risk and demand higher returns to bear it. The problem that several econometric works have shown is that the size of the equity risk premium is too high to be rationally justified. Despite this higher than rationally justifiable premium, there is a strong preference among investors for bonds vs. equity. Even considering indirect holdings (via pension funds or IRAs), only about 37% of American households hold stocks. And this has not changed even after the defined benefit pension plans have started to fall out of use. The Federal Reserve in a 2014 survey reported that the total participation in the stock market (direct and indirect) had fallen from 2007 to 2013. The Federal Reserve estimated that the overall participation is 48.8% and only about 14% of American households hold stocks directly.

This has created a **stock market participation puzzle**, i.e, why are American households failing to take advantage of the equity risk premium and the usually low correlation with other household

risks? And it is not possible to dismiss the puzzle based on education or wealth, for other studies have highlighted that the participation in equities is even lower among wealthier investors (80th percentile of wealth or higher) at only around 10%.

To make all this more surprising still is the fact that there have been clear warnings about individuals' lack of preparedness for retirement in recent times.

Behavioral finance has pointed out that investors have a tendency to check the performance of their portfolios too frequently. With equity investments, doing this causes them to experience many days of losses. If they were to check their investments' performance over longer periods of time, losses would be less frequent. Now, prospect theory teaches us that investors feel the pain of losses far greater than they feel the pleasure of gains. Therefore, the frequent checking of results is a cause of pain and even panic.

Behaviorists have created the term **myopic loss aversion** to describe this combination of short time horizon and framing bias. How can myopic loss aversion impact investment results? Well, investors who check on the values of their portfolios with great frequency are more likely to suffer from this particular bias. And, this is arguably even more so in the internet age as investors are able to check their portfolios in real time or nearly in real time and are thus subjecting themselves to the pain and panic of losses. The end result is that, reacting to these frequent losses, their behavior is to stray from sound investment plans, and they are even more likely to do so in bear markets where the frequency in experiencing losses can make investors very susceptible to frequent trades with a high risk of ending up buying high and selling low, which is the opposite of ideal.

This explanation can also serve to justify the existence of the equity risk premium puzzle: Myopic loss aversion causes investors to demand higher risk premiums to hold on to stocks that could indeed go up but also down, causing said investors to experience pain. In this instance, the functional forms of the value function or the weighting function are not the critical elements; the key factor at work is some sort of psychological cost that stems from the fact that individuals are using the wrong time horizon to evaluate their investments and that pushes them to behave in the way they do.

5. Conclusion

The lesson analyzed prospect theory in details by presenting the two different main phases of editing and evaluation. Then, prospect theory is used to explain a few paradoxes or puzzles that trouble traditional finance.

Once we accept that traditional finance and assuming that economic agents are completely rational do not represent the behavior of real people, the question becomes how can they achieve their financial goals? If an investor displays behavioral biases and uses prospect theory instead of being fully rational in the sense of traditional finance, how do we help them? Behavioral finance suggests that we embrace a mix of adapting and moderating the investor's biases as that would ultimately lead to a better overall result than telling them how to invest. The way to do this is called behaviorally modified asset allocation (BMAA), and it will be covered in the next lesson.

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